

Design Document

Ahmed Furkhan - Personal Portfolio Website

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1 Project Description

Overview

This project is a personal portfolio homepage for Ahmed Furkhan — an AI/ML Engineer and Data Scientist currently pursuing a Master of Science in Computer Science at Northeastern University (Khoury College). The site presents Ahmed's professional identity to three primary audiences: technical recruiters evaluating him for Fall 2026 co-op roles, research collaborators exploring shared interests, and fellow students seeking inspiration or networking.

The site is built entirely with vanilla HTML5, CSS3, and ES6+ modules. There is no jQuery, no component library, and no build step — every feature is implemented in standard web APIs. All JavaScript is split across four ES6 modules (app.js, hexGrid.js, main.js, util.js) and loaded with type="module".

Site Architecture

Page	URL	Key Features
About Me	<code>index.html</code>	Typing animation · Stats row (5 KPI cards) · Featured project grid · AI chatbot FAB
Education	<code>generic.html</code>	Edu-card layout · Northeastern MS + VTU BE with dates
Work Experience	<code>work_experience.html</code>	Vertical timeline · 3 roles with production metrics
Projects ★	<code>elements.html</code>	Canvas hex grid (creative component) + accessible card fallback
Skills	<code>basic.html</code>	Category filter buttons · Flexbox icon grid · 7 categories
Certifications	<code>certify.html</code>	Issuer badge pills · Oracle · Microsoft · Google · HackerRank
AI-Generated	<code>generated.html</code>	Claude-authored narrative · GenAI disclosure — fulfils rubric
Thank You	<code>thank-you.html</code>	Post-form confirmation page

Technology Stack

- HTML5 — Semantic markup, W3C compliant, lang attributes, ARIA labels on all interactive elements
- CSS3 — Flexbox layout, custom properties (CSS variables), @keyframes animations, zero !important
- ES6+ Modules — import/export, arrow functions, async/await, IntersectionObserver, Canvas API
- ESLint + Prettier — linting and formatting enforced before submission
- GitHub Pages — static deployment, no backend, no build step required

2 User Stories

Three user stories cover the primary audience segments. Each story is written in first person and concludes with acceptance criteria that tie back directly to rubric requirements.

Story 1 — Sarah Chen, Senior Technical Recruiter

Role: Senior Technical Recruiter at a Series-B AI startup, Boston, MA

Goal: Evaluate 20+ portfolios per week to fill Fall 2026 AI/ML co-op roles

Context: Sarah receives 10 browser tabs of portfolio links from a Boolean search and opens them simultaneously

The Story

Sarah lands on Ahmed's homepage. The typing animation cycles through 'AI/ML Engineer', 'LLM Evaluation', 'Full-Stack Developer' catches her eye. Below the profile photo she sees five stat cards: 25% LLM quality lift, 85% ML accuracy, 4x reporting cadence. Numbers. Production numbers. She leans forward.

She clicks 'View My Resume'. A clean LaTeX PDF opens in a new tab. She scans the experience section: Outlier (Scale AI), 10 months, LLM evaluation. Feyn Labs, 3 months, clinical ML. Forest Fragrance, data operations. She tabs back and clicks the Projects page. The hex grid loads — she hovers over a tile and it glows blue. She clicks the RAG chatbot hex. It opens. It works.

She fills in the contact details for name, her work email, a short note — and submits. The thank-you page loads instantly. She marks Ahmed as Priority 1 in her ATS.

Acceptance Criteria

- Name, role, and key metrics are visible within 5 seconds of landing (stats row above the fold)
- 'View My Resume' opens the PDF in a new tab without navigating away from the portfolio
- The contact form validates email format and submits without a page reload error
- The hex grid loads and all project tiles open working live applications on click
- The page is fully usable on a 375 px mobile viewport (iPhone) no horizontal scroll

Story 2 — Dr. Marcus Webb, ML Research Scientist

Role: ML Research Scientist at an AI lab, San Francisco, CA

Goal: Find a research assistant or co-author with real LLM alignment experience

Context: A colleague mentions the portfolio at a conference; Marcus Googles it later that evening

The Story

Marcus skips the homepage hero — he's not here for aesthetics. He clicks directly to 'Work Experience'. He reads the Outlier entry carefully: 'designed LLM evaluation workflows', 'structured alignment and safety feedback that directly informed fine-tuning cycles'. That phrase — alignment feedback feeding fine-tuning — tells him this wasn't passive annotation work.

He clicks 'AI-Generated Profile' in the pagination. He reads the narrative and checks the GenAI disclosure — the prompt, the model, the fact that the output was reviewed for accuracy. He approves of the transparency. He clicks the GitHub icon in the nav. Three AI repos, all public, all with READMEs.

He emails Ahmed directly: 'Collaboration interest — LLM evaluation survey'.

Acceptance Criteria

- Work Experience section provides technical depth — methodology at Outlier, not just job title
- The generated.html AI-generated profile clearly discloses the GenAI model and prompt used
- GitHub link in nav icons opens Ahmed's public repositories
- All six project cards distinguish AI/ML system work from simple web demos
- Page loads without layout shift or broken images on a desktop at 1280 px wide

Story 3 — Priya Sharma, MS CS Student

Role: First-year MS Computer Science student, Northeastern University, Boston, MA

Goal: Get inspiration for her own portfolio and understand how the hex grid was built

Context: A classmate shares the portfolio link in the Khoury Slack channel with the message: 'This is the bar'

The Story

Priya clicks straight to Projects. She sees a dark hexagonal grid. She hovers over the first tile — it glows blue and a small arrow appears. She clicks. The RAG chatbot opens in a new tab. She right-clicks the page and opens DevTools. She finds hexGrid.js in the Sources panel and starts reading the flat-top hex geometry, the image clipping, the event listeners. She takes notes in her Notion for twenty minutes.

She navigates to Skills and clicks the 'AI/ML' filter. The other sections disappear instantly. She clicks 'Databases'. Only MySQL, MongoDB, and SQLite remain. She clicks 'All'. Everything returns without a page reload. Clean.

She follows Ahmed on LinkedIn and messages him: 'I'm a first-year MS student at Khoury. Love the hex grid. Would you be open to a coffee chat?'

Acceptance Criteria

- Canvas hex grid renders within 2 seconds of page load on a standard broadband connection
- Hovering a hexagon triggers a visible blue glow and border change (CSS + Canvas redraw)
- Clicking a hexagon opens the correct live project URL in a new tab
- The skill category filter buttons hide/show sections without a page reload
- All skill logo images have descriptive alt text (not empty alt="" or generic 'logo')

3 Design Mockups (Wireframes)

All wireframes use consistent visual language: dark fills for chrome and navigation bars, light grey for content containers, cyan accent for interactive elements and active states, and dashed borders for optional/toggleable components. Dimensions are proportionally accurate to the final implementation.

Wireframe 1 — Homepage (index.html) · Desktop 1280 px

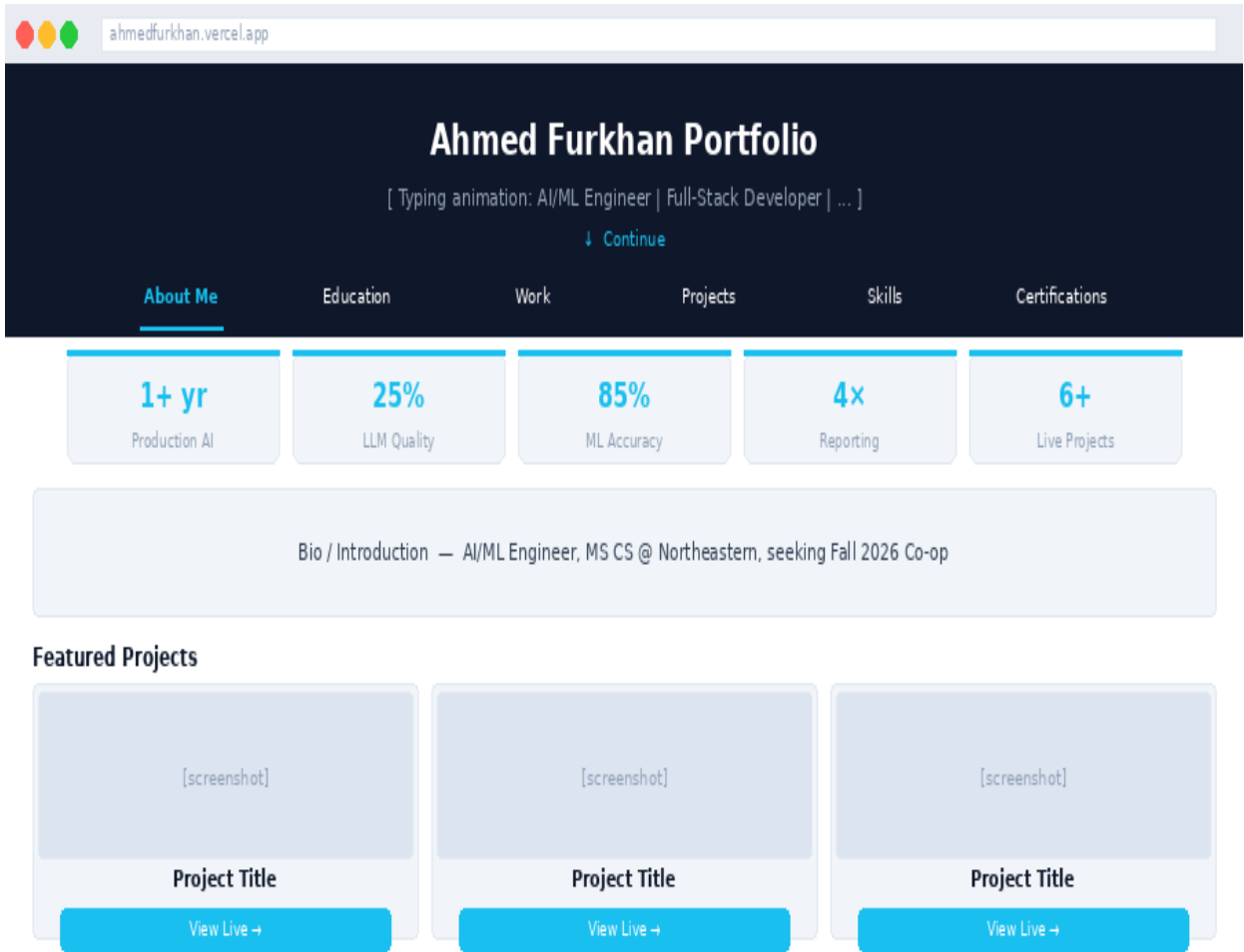


Fig 1. Homepage showing hero intro, sticky nav, stats row (5 KPI cards), bio block, and 3-column featured project grid. Dialogflow chatbot FAB visible bottom-right.

Wireframe 2 — Projects Page (elements.html) Desktop 1280 px

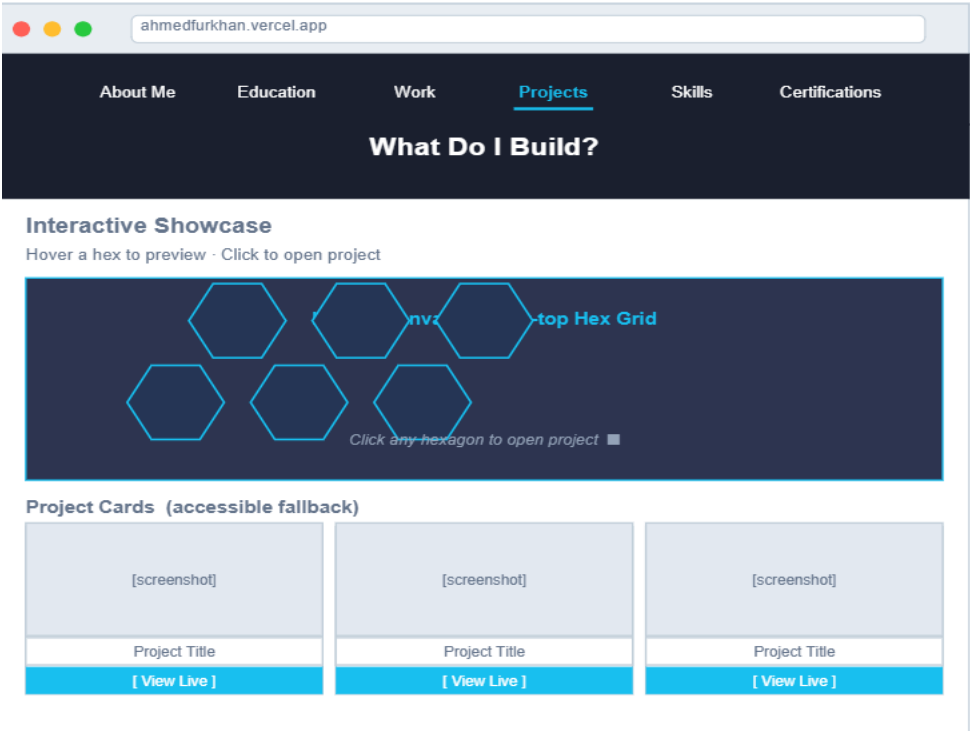


Fig 2. Projects page — HTML5 Canvas hex grid at top (creative original component) with accessible card grid below as semantic fallback for screen-readers.

Wireframe 3 — Work Experience (work_experience.html) Desktop

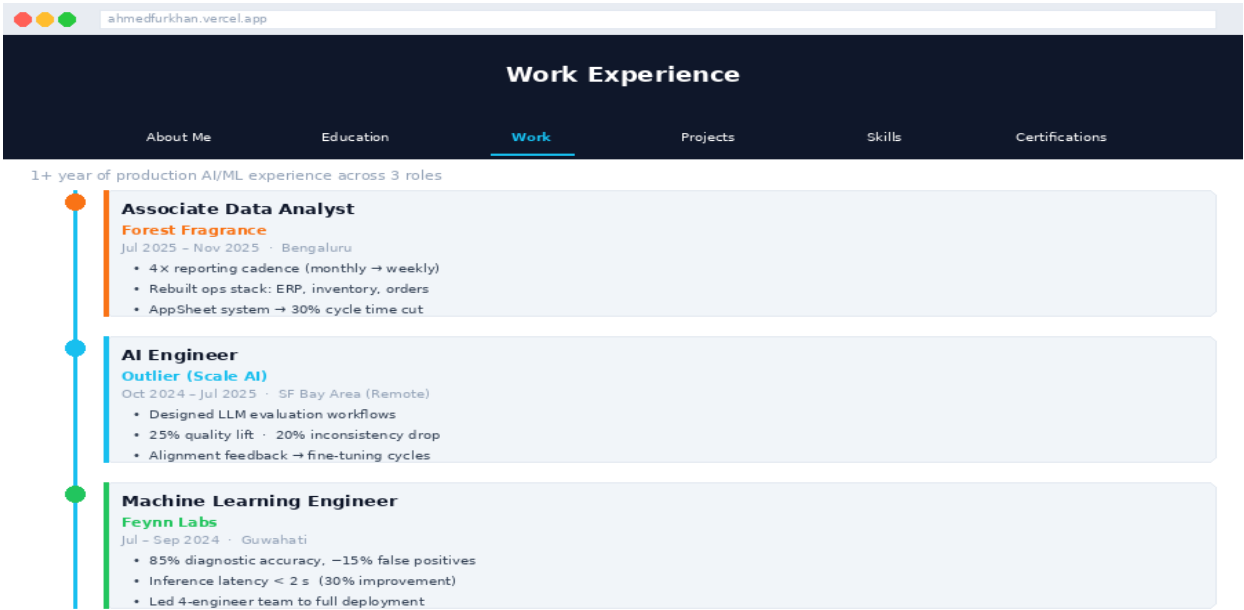


Fig 3. Vertical timeline with gradient spine. Each role has a colored dot (orange = Forest Fragrance, cyan = Outlier, green = Feynn Labs). Newest role at top.

Wireframe 4 — Skills Page (basic.html) · Desktop

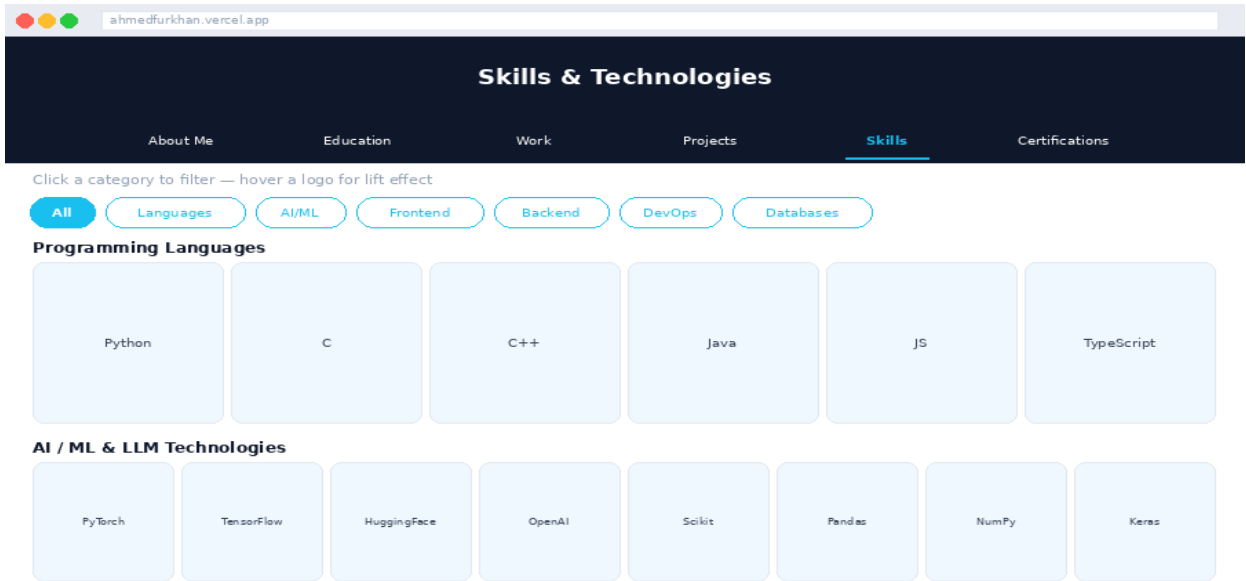


Fig 4. Filter bar at top — clicking a category instantly hides all other sections via app.js `initSkillFilter()`. 'All' pill is active-state cyan by default. Icon grid uses CSS Flexbox with wrap.

Wireframe 5 — Mobile Layout · 375 px viewport

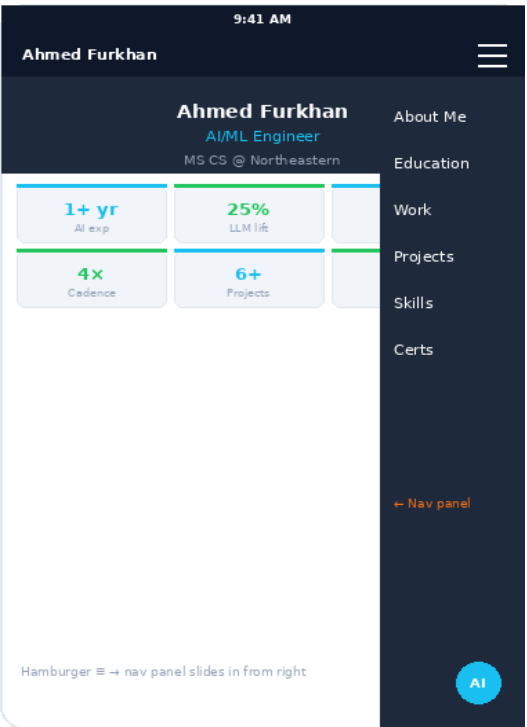


Fig 5. Mobile — hamburger (≡) replaces nav. Slide-out #navPanel from right. Stats in 3×2 grid. Chatbot FAB fixed bottom-right.

Mobile design notes

- Hamburger (≡) top-right triggers the #navPanel slide-out drawer (built in vanilla JS, no jQuery)
- Stats row reorganises from 5-wide to 3×2 grid using CSS flex-wrap
- All touch targets are ≥ 44 px high (iOS HIG minimum)
- Chatbot FAB stays fixed at bottom-right on all viewport sizes
- Font sizes scale down but remain readable — minimum 14 px for body copy
- Hex grid canvas recalculates column count on resize — shrinks from 3-wide to 2-wide on mobile

Responsive breakpoints

≥ 981 px	Full nav bar visible
737–980 px	Hamburger replaces nav
≤ 736 px	Stats 3×2, cards stack

Wireframe 6 — Education (generic.html) · Desktop

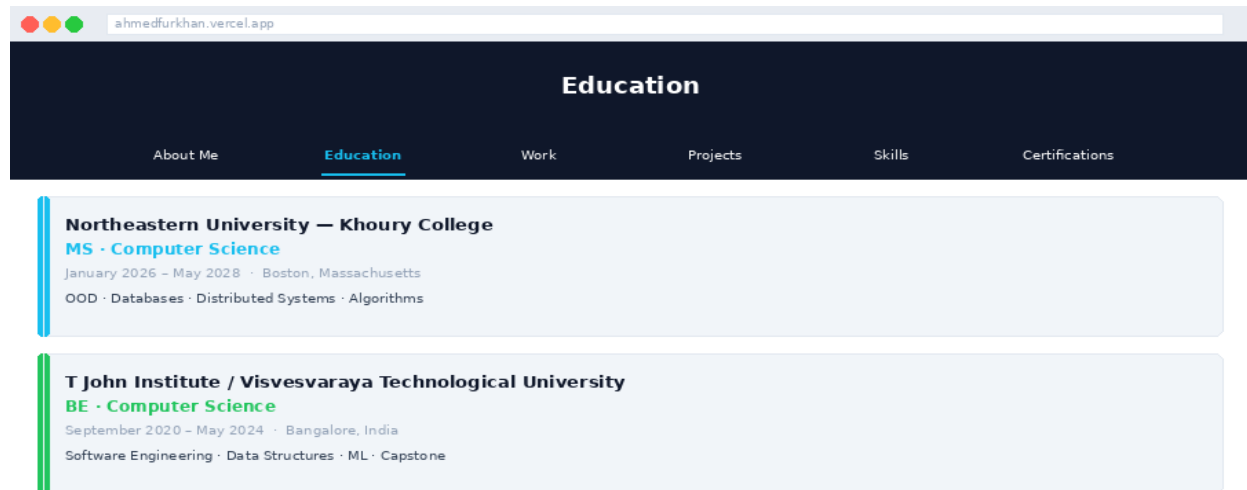


Fig 6. Education page using Edu-card layout with colored left border per institution. Northeastern MS card is primary (cyan); VTU BE card is secondary (green).